

Table 1: NanoAOD 2025 datasets used in the analysis

| Sample                   | NanoAOD dataset                             |
|--------------------------|---------------------------------------------|
| <b>Data (96 samples)</b> |                                             |
| Muon0_Run2025B_v1        | /Muon0/Run2025B-PromptReco-v1/NANOAOD       |
| Muon0_Run2025C_v1        | /Muon0/Run2025C-PromptReco-v1/NANOAOD       |
| Muon0_Run2025C_v2        | /Muon0/Run2025C-PromptReco-v2/NANOAOD       |
| Muon0_Run2025D_v1        | /Muon0/Run2025D-PromptReco-v1/NANOAOD       |
| Muon0_Run2025E_v1        | /Muon0/Run2025E-PromptReco-v1/NANOAOD       |
| Muon0_Run2025F_v1        | /Muon0/Run2025F-PromptReco-v1/NANOAOD       |
| Muon0_Run2025F_v2        | /Muon0/Run2025F-PromptReco-v2/NANOAOD       |
| Muon0_Run2025G_v1        | /Muon0/Run2025G-PromptReco-v1/NANOAOD       |
| Muon1_Run2025B_v1        | /Muon1/Run2025B-PromptReco-v1/NANOAOD       |
| Muon1_Run2025C_v1        | /Muon1/Run2025C-PromptReco-v1/NANOAOD       |
| Muon1_Run2025C_v2        | /Muon1/Run2025C-PromptReco-v2/NANOAOD       |
| Muon1_Run2025D_v1        | /Muon1/Run2025D-PromptReco-v1/NANOAOD       |
| Muon1_Run2025E_v1        | /Muon1/Run2025E-PromptReco-v1/NANOAOD       |
| Muon1_Run2025F_v1        | /Muon1/Run2025F-PromptReco-v1/NANOAOD       |
| Muon1_Run2025F_v2        | /Muon1/Run2025F-PromptReco-v2/NANOAOD       |
| Muon1_Run2025G_v1        | /Muon1/Run2025G-PromptReco-v1/NANOAOD       |
| MuonEG_Run2025B_v1       | /MuonEG/Run2025B-PromptReco-v1/NANOAOD      |
| MuonEG_Run2025C_v1       | /MuonEG/Run2025C-PromptReco-v1/NANOAOD      |
| MuonEG_Run2025C_v2       | /MuonEG/Run2025C-PromptReco-v2/NANOAOD      |
| MuonEG_Run2025D_v1       | /MuonEG/Run2025D-PromptReco-v1/NANOAOD      |
| MuonEG_Run2025E_v1       | /MuonEG/Run2025E-PromptReco-v1/NANOAOD      |
| MuonEG_Run2025F_v1       | /MuonEG/Run2025F-PromptReco-v1/NANOAOD      |
| MuonEG_Run2025F_v2       | /MuonEG/Run2025F-PromptReco-v2/NANOAOD      |
| MuonEG_Run2025G_v1       | /MuonEG/Run2025G-PromptReco-v1/NANOAOD      |
| ParkingVBF0_Run2025B_v1  | /ParkingVBF0/Run2025B-PromptReco-v1/NANOAOD |
| ParkingVBF0_Run2025C_v1  | /ParkingVBF0/Run2025C-PromptReco-v1/NANOAOD |
| ParkingVBF0_Run2025C_v2  | /ParkingVBF0/Run2025C-PromptReco-v2/NANOAOD |
| ParkingVBF0_Run2025D_v1  | /ParkingVBF0/Run2025D-PromptReco-v1/NANOAOD |
| ParkingVBF0_Run2025E_v1  | /ParkingVBF0/Run2025E-PromptReco-v1/NANOAOD |
| ParkingVBF0_Run2025F_v1  | /ParkingVBF0/Run2025F-PromptReco-v1/NANOAOD |
| ParkingVBF0_Run2025F_v2  | /ParkingVBF0/Run2025F-PromptReco-v2/NANOAOD |
| ParkingVBF0_Run2025G_v1  | /ParkingVBF0/Run2025G-PromptReco-v1/NANOAOD |
| ParkingVBF1_Run2025B_v1  | /ParkingVBF1/Run2025B-PromptReco-v1/NANOAOD |
| ParkingVBF1_Run2025C_v1  | /ParkingVBF1/Run2025C-PromptReco-v1/NANOAOD |
| ParkingVBF1_Run2025C_v2  | /ParkingVBF1/Run2025C-PromptReco-v2/NANOAOD |
| ParkingVBF1_Run2025D_v1  | /ParkingVBF1/Run2025D-PromptReco-v1/NANOAOD |
| ParkingVBF1_Run2025E_v1  | /ParkingVBF1/Run2025E-PromptReco-v1/NANOAOD |
| ParkingVBF1_Run2025F_v1  | /ParkingVBF1/Run2025F-PromptReco-v1/NANOAOD |
| ParkingVBF1_Run2025F_v2  | /ParkingVBF1/Run2025F-PromptReco-v2/NANOAOD |
| ParkingVBF1_Run2025G_v1  | /ParkingVBF1/Run2025G-PromptReco-v1/NANOAOD |
| ParkingVBF2_Run2025B_v1  | /ParkingVBF2/Run2025B-PromptReco-v1/NANOAOD |
| ParkingVBF2_Run2025C_v1  | /ParkingVBF2/Run2025C-PromptReco-v1/NANOAOD |
| ParkingVBF2_Run2025C_v2  | /ParkingVBF2/Run2025C-PromptReco-v2/NANOAOD |
| ParkingVBF2_Run2025D_v1  | /ParkingVBF2/Run2025D-PromptReco-v1/NANOAOD |
| ParkingVBF2_Run2025E_v1  | /ParkingVBF2/Run2025E-PromptReco-v1/NANOAOD |
| ParkingVBF2_Run2025F_v1  | /ParkingVBF2/Run2025F-PromptReco-v1/NANOAOD |
| ParkingVBF2_Run2025F_v2  | /ParkingVBF2/Run2025F-PromptReco-v2/NANOAOD |
| ParkingVBF2_Run2025G_v1  | /ParkingVBF2/Run2025G-PromptReco-v1/NANOAOD |
| ParkingVBF3_Run2025B_v1  | /ParkingVBF3/Run2025B-PromptReco-v1/NANOAOD |
| ParkingVBF3_Run2025C_v1  | /ParkingVBF3/Run2025C-PromptReco-v1/NANOAOD |
| ParkingVBF3_Run2025C_v2  | /ParkingVBF3/Run2025C-PromptReco-v2/NANOAOD |
| ParkingVBF3_Run2025D_v1  | /ParkingVBF3/Run2025D-PromptReco-v1/NANOAOD |
| ParkingVBF3_Run2025E_v1  | /ParkingVBF3/Run2025E-PromptReco-v1/NANOAOD |

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Table 1: NanoAOD 2025 datasets used in the analysis (continued)

| Sample                          | NanoAOD dataset                                                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ParkingVBF3_Run2025F_v1         | /ParkingVBF3/Run2025F-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF3_Run2025F_v2         | /ParkingVBF3/Run2025F-PromptReco-v2/NANOAOD                                                                                                     |
| ParkingVBF3_Run2025G_v1         | /ParkingVBF3/Run2025G-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF4_Run2025B_v1         | /ParkingVBF4/Run2025B-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF4_Run2025C_v1         | /ParkingVBF4/Run2025C-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF4_Run2025C_v2         | /ParkingVBF4/Run2025C-PromptReco-v2/NANOAOD                                                                                                     |
| ParkingVBF4_Run2025D_v1         | /ParkingVBF4/Run2025D-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF4_Run2025E_v1         | /ParkingVBF4/Run2025E-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF4_Run2025F_v1         | /ParkingVBF4/Run2025F-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF4_Run2025F_v2         | /ParkingVBF4/Run2025F-PromptReco-v2/NANOAOD                                                                                                     |
| ParkingVBF4_Run2025G_v1         | /ParkingVBF4/Run2025G-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF5_Run2025B_v1         | /ParkingVBF5/Run2025B-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF5_Run2025C_v1         | /ParkingVBF5/Run2025C-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF5_Run2025C_v2         | /ParkingVBF5/Run2025C-PromptReco-v2/NANOAOD                                                                                                     |
| ParkingVBF5_Run2025D_v1         | /ParkingVBF5/Run2025D-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF5_Run2025E_v1         | /ParkingVBF5/Run2025E-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF5_Run2025F_v1         | /ParkingVBF5/Run2025F-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF5_Run2025F_v2         | /ParkingVBF5/Run2025F-PromptReco-v2/NANOAOD                                                                                                     |
| ParkingVBF5_Run2025G_v1         | /ParkingVBF5/Run2025G-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF6_Run2025B_v1         | /ParkingVBF6/Run2025B-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF6_Run2025C_v1         | /ParkingVBF6/Run2025C-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF6_Run2025C_v2         | /ParkingVBF6/Run2025C-PromptReco-v2/NANOAOD                                                                                                     |
| ParkingVBF6_Run2025D_v1         | /ParkingVBF6/Run2025D-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF6_Run2025E_v1         | /ParkingVBF6/Run2025E-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF6_Run2025F_v1         | /ParkingVBF6/Run2025F-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF6_Run2025F_v2         | /ParkingVBF6/Run2025F-PromptReco-v2/NANOAOD                                                                                                     |
| ParkingVBF6_Run2025G_v1         | /ParkingVBF6/Run2025G-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF7_Run2025B_v1         | /ParkingVBF7/Run2025B-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF7_Run2025C_v1         | /ParkingVBF7/Run2025C-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF7_Run2025C_v2         | /ParkingVBF7/Run2025C-PromptReco-v2/NANOAOD                                                                                                     |
| ParkingVBF7_Run2025D_v1         | /ParkingVBF7/Run2025D-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF7_Run2025E_v1         | /ParkingVBF7/Run2025E-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF7_Run2025F_v1         | /ParkingVBF7/Run2025F-PromptReco-v1/NANOAOD                                                                                                     |
| ParkingVBF7_Run2025F_v2         | /ParkingVBF7/Run2025F-PromptReco-v2/NANOAOD                                                                                                     |
| ParkingVBF7_Run2025G_v1         | /ParkingVBF7/Run2025G-PromptReco-v1/NANOAOD                                                                                                     |
| Tau_Run2025B_v1                 | /Tau/Run2025B-PromptReco-v1/NANOAOD                                                                                                             |
| Tau_Run2025C_v1                 | /Tau/Run2025C-PromptReco-v1/NANOAOD                                                                                                             |
| Tau_Run2025C_v2                 | /Tau/Run2025C-PromptReco-v2/NANOAOD                                                                                                             |
| Tau_Run2025D_v1                 | /Tau/Run2025D-PromptReco-v1/NANOAOD                                                                                                             |
| Tau_Run2025E_v1                 | /Tau/Run2025E-PromptReco-v1/NANOAOD                                                                                                             |
| Tau_Run2025F_v1                 | /Tau/Run2025F-PromptReco-v1/NANOAOD                                                                                                             |
| Tau_Run2025F_v2                 | /Tau/Run2025F-PromptReco-v2/NANOAOD                                                                                                             |
| Tau_Run2025G_v1                 | /Tau/Run2025G-PromptReco-v1/NANOAOD                                                                                                             |
| <b>Simulation (177 samples)</b> |                                                                                                                                                 |
| GluGluHto2Mu_M120               | /GluGluH-Hto2Mu_Par-M-120_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM          |
| GluGluHto2Mu                    | /GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                |
|                                 | /GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM           |
| GluGluHto2Mu_amcatnlo           | /GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM          |
|                                 | /GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM     |
| GluGluHto2Mu_MinNLO             | /GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM          |
| GluGluHto2Mu_tuneDown           | /GluGluH-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM      |
|                                 | /GluGluH-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM |
| GluGluHto2Mu_tuneUp             | /GluGluH-Hto2Mu_Par-M-125_TuneCP5Up_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM        |
| GluGluHto2Mu_M130               | /GluGluH-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM          |
|                                 | /GluGluH-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM     |

Table 1: NanoAOD 2025 datasets used in the analysis (continued)

| Sample                                    | NanoAOD dataset                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VBFHto2Mu_M125_powhcg                     | /VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM<br>/VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM                                                                                                                                                                                                 |
| VBFHto2Mu_M120                            | /VBF-Hto2Mu_Par-M-120_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                      |
| VBFHto2Mu_m125_tuneCP5Up_amcatnlo         | /VBF-Hto2Mu_Par-M-125_TuneCP5Up_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                    |
| VBFHto2Mu_m125_tuneCP5Down_amcatnlo       | /VBF-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM<br>/VBF-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM                                                                                                                                                                                       |
| VBFHto2Mu_M125_amcatnlo                   | /VBF-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM<br>/VBF-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM                                                                                                                                                                                               |
| VBFHto2Mu_M130                            | /VBF-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM<br>/VBF-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM                                                                                                                                                                                               |
| ggZH_Hto2Mu_ZtoAll_M125                   | /ggZH_HtoMuMu_ZToAll_M125_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                    |
| VBFHto2Mu_m125_Flashsim                   | /VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhcg-pythia8/fvaselli-VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhcg-pythia8-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER<br>/VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhcg-pythia8/fvaselli-VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhcg-pythia8-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2_ext1-v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER |
| ZH_Hto2Mu                                 | /ZH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhcgMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                                                                                                                                                                                                                                                                                                    |
| WplusH_Hto2Mu                             | /WplusH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhcgMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                                                                                                                                                                                                                                                                                                |
| WminusH_Hto2Mu                            | /WminusH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhcgMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                                                                                                                                                                                                                                                                                               |
| TTH_Hto2Mu                                | /TTH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM<br>/TTH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM                                                                                                                                                                                                   |
| DYto2Mu_M_50_OJ_amcatnloFFFX              | /DYto2Mu-2Jets_Bin-OJ-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                                                                                                                                                                                                                                                                                           |
| DYto2Mu_M_50_1J_amcatnloFFFX              | /DYto2Mu-2Jets_Bin-1J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v1/NANOADSIM                                                                                                                                                                                                                                                                                                                           |
| DYto2Mu_M_50_2J_amcatnloFFFX              | /DYto2Mu-2Jets_Bin-2J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                                                                                                                                                                                                                                                                                           |
| DYto2Mu_M_10to50_amcatnloFFFX             | /DYto2Mu-2Jets_Bin-MLL-10to50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                          |
| DYto2Mu_M_50_amcatnloFFFX                 | /DYto2Mu-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v6/NANOADSIM                                                                                                                                                                                                                                                                                                                              |
| DYto2Mu_MLL_105to160_amcatnloFFFX         | /DYto2Mu-2Jets_Bin-MLL-105to160_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM<br>/DYto2Mu-2Jets_Bin-MLL-105to160_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM                                                                                                                                                                   |
| DYto2Mu_MLL_105to160_amcatnloFFFX_Fil_VBF | /DYto2Mu-2Jets_Bin-MLL-105to160_Fil-VBF_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                |
| DYto2E_M_50_OJ_amcatnloFFFX               | /DYto2E-2Jets_Bin-OJ-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                            |
| DYto2E_M_50_1J_amcatnloFFFX               | /DYto2E-2Jets_Bin-1J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                                                                                                                                                                                                                                                                                            |
| DYto2E_M_50_2J_amcatnloFFFX               | /DYto2E-2Jets_Bin-2J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                            |
| DYto2E_M_10to50_amcatnloFFFX              | /DYto2E-2Jets_Bin-MLL-10to50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                           |
| DYto2E_M_50_amcatnloFFFX                  | /DYto2E-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v4/NANOADSIM                                                                                                                                                                                                                                                                                                                               |
| DYto2Tau_M_50_OJ_amcatnloFFFX             | /DYto2Tau-2Jets_Bin-OJ-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v4/NANOADSIM                                                                                                                                                                                                                                                                                                                          |
| DYto2Tau_M_50_1J_amcatnloFFFX             | /DYto2Tau-2Jets_Bin-1J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                          |
| DYto2Tau_M_50_2J_amcatnloFFFX             | /DYto2Tau-2Jets_Bin-2J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                          |
| DYto2Tau_M_10to50_amcatnloFFFX            | /DYto2Tau-2Jets_Bin-MLL-10to50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                         |
| DYto2Tau_M_50_amcatnloFFFX                | /DYto2Tau-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v7/NANOADSIM                                                                                                                                                                                                                                                                                                                             |
| DYto2Mu_MLL_10to50_powhcg                 | /DYto2Mu_Bin-MLL-10to50_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                      |
| DYto2Mu_MLL_50to120_powhcg                | /DYto2Mu_Bin-MLL-50to120_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                     |
| DYto2Mu_MLL_120to200_powhcg               | /DYto2Mu_Bin-MLL-120to200_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                    |
| DYto2Mu_MLL_200to400_powhcg               | /DYto2Mu_Bin-MLL-200to400_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                    |
| DYto2Mu_MLL_400to800_powhcg               | /DYto2Mu_Bin-MLL-400to800_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                    |
| DYto2Mu_MLL_800to1500_powhcg              | /DYto2Mu_Bin-MLL-800to1500_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                   |
| DYto2Mu_MLL_1500to2500_powhcg             | /DYto2Mu_Bin-MLL-1500to2500_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                  |
| DYto2Mu_MLL_2500to4000_powhcg             | /DYto2Mu_Bin-MLL-2500to4000_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                  |
| DYto2Mu_MLL_4000to6000_powhcg             | /DYto2Mu_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                  |
| DYto2Mu_MLL_6000_powhcg                   | /DYto2Mu_Bin-MLL-6000_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                        |
| DYto2Tau_MLL_10to50_powhcg                | /DYto2Tau_Bin-MLL-10to50_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                     |
| DYto2Tau_MLL_50to120_powhcg               | /DYto2Tau_Bin-MLL-50to120_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                    |
| DYto2Tau_MLL_120to200_powhcg              | /DYto2Tau_Bin-MLL-120to200_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                   |
| DYto2Tau_MLL_200to400_powhcg              | /DYto2Tau_Bin-MLL-200to400_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                   |
| DYto2Tau_MLL_400to800_powhcg              | /DYto2Tau_Bin-MLL-400to800_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                   |
| DYto2Tau_MLL_800to1500_powhcg             | /DYto2Tau_Bin-MLL-800to1500_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                  |
| DYto2Tau_MLL_1500to2500_powhcg            | /DYto2Tau_Bin-MLL-1500to2500_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                 |
| DYto2Tau_MLL_2500to4000_powhcg            | /DYto2Tau_Bin-MLL-2500to4000_TuneCP5_13p6TeV_powhcg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                                                                                                                                                                                                                 |

Continued on next page

Table 1: NanoAOD 2025 datasets used in the analysis (continued)

| Sample                                     | NanoAOD dataset                                                                                                                                                                                                                                        |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DYto2Tau_MLL_4000to6000_powheg             | /DYto2Tau_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                    |
| DYto2Tau_MLL_6000_powheg                   | /DYto2Tau_Bin-MLL-6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                          |
| DYto2E_MLL_10to50_powheg                   | /DYto2E_Bin-MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                          |
| DYto2E_MLL_50to120_powheg                  | /DYto2E_Bin-MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                         |
| DYto2E_MLL_120to200_powheg                 | /DYto2E_Bin-MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                        |
| DYto2E_MLL_200to400_powheg                 | /DYto2E_Bin-MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                        |
| DYto2E_MLL_400to800_powheg                 | /DYto2E_Bin-MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                        |
| DYto2E_MLL_800to1500_powheg                | /DYto2E_Bin-MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                       |
| DYto2E_MLL_1500to2500_powheg               | /DYto2E_Bin-MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                      |
| DYto2E_MLL_2500to4000_powheg               | /DYto2E_Bin-MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                      |
| DYto2E_MLL_4000to6000_powheg               | /DYto2E_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                      |
| DYto2E_MLL_6000_powheg                     | /DYto2E_Bin-MLL-6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                            |
| DYto2Mu_MLL_50to130_powheg_minnlo          | /DYto2Mu_Bin-MLL-50to130_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                           |
| DYto2Mu_MLL_130to200_powheg_minnlo         | /DYto2Mu_Bin-MLL-130to200_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                          |
| DYto2Mu_MLL_200to400_powheg_minnlo         | /DYto2Mu_Bin-MLL-200to400_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                          |
| DYto2Mu_MLL_400to600_powheg_minnlo         | /DYto2Mu_Bin-MLL-400to600_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                          |
| DYto2Mu_MLL_600to800_powheg_minnlo         | /DYto2Mu_Bin-MLL-600to800_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                          |
| DYto2Mu_MLL_1000to1500_powheg_minnlo       | /DYto2Mu_Bin-MLL-1000to1500_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                        |
| DYto2Mu_MLL_1500to2000_powheg_minnlo       | /DYto2Mu_Bin-MLL-1500to2000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                        |
| DYto2Mu_MLL_2000to4000_powheg_minnlo       | /DYto2Mu_Bin-MLL-2000to4000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                        |
| DYto2Mu_MLL_4000to6000_powheg_minnlo       | /DYto2Mu_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                        |
| DYto2Mu_MLL_6000to13600_powheg_minnlo      | /DYto2Mu_Bin-MLL-6000to13600_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                       |
| DYto2L_M_50_amcatnloFXXF_Flashsim          | /DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXXF-pythia8/fvaselli-flashsim_DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXXF-pythia8-Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic-937767ad549695a10e40fd392f6c3633/USER                     |
| DYto2Mu_MLL_105to160_amcatnloFXXF_Flashsim | /DYto2Mu-2Jets_Bin-MLL-105to160_TuneCP5_13p6TeV_amcatnloFXXF-pythia8/fvaselli-DY2Mu_2Jets_MLL-105to160-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-packedTrackGenJetAK4-limit-TrkGJ_eta-to-tracker-66034083b387d860d393385d18a4f59d/USER |
| DYto2Mu_M_50_amcatnloFXXF_Flashsim         | /DYto2Mu-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXXF-pythia8/fvaselli-SoftActivity_NanoAOD_Test_v1-ace5d3f4ddab40d8fc0f18e7a8dbab44/USER                                                                                                             |
| EWK_2L2J_madgraph_herwig                   | /EWK-2L2J_Bin-MLL-50-MJJ-120_TuneCH3_13p6TeV_madgraph-herwig7/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                  |
|                                            | /EWK-2L2J_Bin-MLL-50-MJJ-120_TuneCH3_13p6TeV_madgraph-herwig7/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM                                                                                                             |
| EWK_2Mu2J_MLL_105to160_herwig              | /EWK-2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                   |
|                                            | /EWK-2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM                                                                                                              |
| EWK_2Mu2J_MLL_105to160_pythia              | /EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                          |
|                                            | /EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM                                                                                                     |
| TTLNu-EWK_amcatnlo                         | /TTLNu-EWK_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                    |
| WmWmJJ_EWK                                 | /WmWmJJ-EWK_TuneCP5_13p6TeV-powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                     |
| EWK_2Mu2J_MLL_105to160_pythia_Flashsim     | /EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/fvaselli-EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER                              |
| ggZH_Hto2B_Zto2Q                           | /GluGluZH-Zto2Q-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                 |
| ggZH_Hto2B_Zto2L                           | /GluGluZH-Zto2L-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                 |
| GluGluHto2B_M125                           | /GluGluH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                   |
|                                            | /GluGluH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM                                                                                                              |
| GluGluHto2Tau_M125                         | /GluGluHto2Tau_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                  |
| GluGluHto2Tau_UnCorrelatedDecay_Filtered   | /GluGluH-Hto2TauUnCorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                       |
| GluGluHto2Tau_UnCorrelatedDecay_UnFiltered | /GluGluH-Hto2TauUnCorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                     |
| GluGluHto2Wto2L2Nu_M125                    | /GluGluHto2Wto2L2Nu_Par-M-125_TuneCP5_13p6TeV_powheg-jhugen-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                            |
| TTHto2B_M125                               | /TTH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                            |
| TTHtoNon2B_M125                            | /TTH-HtoNon2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                         |
| TTZH_ZHto4B                                | /TTZH-ZHto4B_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                                  |
| VBFHto2B_M125                              | /VBFH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                           |
|                                            | /VBFH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM                                                                                                                      |
| VBFHto2Tau_M125                            | /VBFH-Hto2Tau_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                                         |
| VBFHto2Tau_UnCorrelatedDecay_UnFiltered    | /VBFH-Hto2TauUnCorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                          |
| VBFHto2Tau_UnCorrelatedDecay_Filtered      | /VBFH-Hto2TauUnCorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                        |
| VBFHto2Wto2L2Nu_M125                       | /VBFHto2Wto2L2Nu_Par-M-125_TuneCP5_13p6TeV_powheg-jhugen-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                               |
| WminusH_Hto2B_WtoLnu                       | /WminusH-WtoLnu-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                            |
| WplusH_Hto2B_WtoLnu                        | /WplusH-WtoLnu-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                             |

Continued on next page

Table 1: NanoAOD 2025 datasets used in the analysis (continued)

| Sample                                     | NanoAOD dataset                                                                                                                                                                                    |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WminusHto2Tau_UncorrelatedDecay_Filtered   | /WminusH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powhegMINLLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                             |
| WminusHto2Tau_UncorrelatedDecay_UnFiltered | /WminusH-Hto2TauUncorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powhegMINLLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                           |
| WplusHto2Tau_UncorrelatedDecay_Filtered    | /WplusH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powhegMINLLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                              |
| WplusHto2Tau_UncorrelatedDecay_UnFiltered  | /WplusH-Hto2TauUncorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powhegMINLLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                            |
| ZH_Hto2B_Zto2L                             | /ZH-Zto2L-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                              |
| ZH_Hto2B_Zto2Q                             | /ZH-Zto2Q-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                              |
| TbarBtoLminusNuB_s_channel_4FS             | /TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                           |
| TBbartoLplusNuBbar_s_channel_4FS           | /TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                         |
| TbarBQtoLnu_t_channel_4FS                  | /TbarBQtoLnu-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                          |
| TbarBQto2Q_t_channel_4FS                   | /TbarBQto2Q-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                           |
| TBbarQtoLnu_t_channel_4FS                  | /TBbarQtoLnu-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                          |
| TBbarQto2Q_t_channel_4FS                   | /TBbarQto2Q-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                           |
| TbarWplusto2L2Nu                           | /TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                           |
| TbarWplusto4Q                              | /TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                              |
| TbarWplustoLnu2Q                           | /TbarWplustoLnu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                           |
| TWminusto2L2Nu                             | /TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                             |
| TWminusto4Q                                | /TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                |
| TWminustoLnu2Q                             | /TWminustoLnu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                             |
| TTto2L2Nu                                  | /TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                                                  |
| TTto4Q                                     | /TTto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                     |
| TTtoLnu2Q                                  | /TTtoLnu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                  |
| TTWH                                       | /TTWH_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                     |
| TTWW                                       | /TTWW_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                     |
| TTWZ                                       | /TTWZ_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                     |
| TTZZ_ZZto4B                                | /TTZZ-ZZto4B_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM/                                                                             |
| TTZ_Zto2Q                                  | /TTZ-ZtoQQ-1Jets_TuneCP5_13p6TeV_amcatnloFXFXold-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                   |
| TTto2L2Nu_Flashsim                         | /TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/fvaselli-TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER |
| TT_Flashsim                                | /TT_TuneCH3_13TeV_powheg-herwig7/fvaselli-flashsim_v9-f5e5c27141f6cbae6ce3c7d60feb1683/USER                                                                                                        |
| WtoLnu_1J_madgraphMLM                      | /WtoLnu-4Jets_Bin-1J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                   |
| WtoLnu_2J_amcatnloFXFX                     | /WtoLnu-4Jets_Bin-2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                   |
| WtoLnu_2J_madgraphMLM                      | /WtoLnu-4Jets_Bin-2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                   |
| WtoLnu_3J_madgraphMLM                      | /WtoLnu-4Jets_Bin-3J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                   |
| WtoLnu_4J_madgraphMLM                      | /WtoLnu-4Jets_Bin-4J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                   |
| WtoLnu_PTLNu_40to100_1J_amcatnloFXFX       | /WtoLnu-2Jets_Bin-1J-PTLNU-40to100_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                    |
| WtoLnu_PTLNu_40to100_2J_amcatnloFXFX       | /WtoLnu-2Jets_Bin-2J-PTLNU-40to100_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                    |
| WtoLnu_PTLNu_100to200_1J_amcatnloFXFX      | /WtoLnu-2Jets_Bin-1J-PTLNU-100to200_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                   |
| WtoLnu_PTLNu_100to200_2J_amcatnloFXFX      | /WtoLnu-2Jets_Bin-2J-PTLNU-100to200_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                   |
| WtoLnu_PTLNu_200to400_1J_amcatnloFXFX      | /WtoLnu-2Jets_Bin-1J-PTLNU-200to400_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                   |
| WtoLnu_PTLNu_200to400_2J_amcatnloFXFX      | /WtoLnu-2Jets_Bin-2J-PTLNU-200to400_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                   |
| WtoLnu_PTLNu_400to600_1J_amcatnloFXFX      | /WtoLnu-2Jets_Bin-1J-PTLNU-400to600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                   |
| WtoLnu_PTLNu_400to600_2J_amcatnloFXFX      | /WtoLnu-2Jets_Bin-2J-PTLNU-400to600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                   |
| WtoLnu_PTLNu_600_1J_amcatnloFXFX           | /WtoLnu-2Jets_Bin-1J-PTLNU-600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                        |
| WtoLnu_PTLNu_600_2J_amcatnloFXFX           | /WtoLnu-2Jets_Bin-2J-PTLNU-600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                        |
| WtoENu_amcatnloFXFX                        | /WtoENu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                                         |
| WtoMuNu_amcatnloFXFX                       | /WtoMuNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                                        |
| WtoTauNu_amcatnloFXFX                      | /WtoTauNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                                                                       |
| WW                                         | /WW_TuneCP5_13p6TeV_pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                |
| WZ                                         | /WZ_TuneCP5_13p6TeV_pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                |
| ZZ                                         | /ZZ_TuneCP5_13p6TeV_pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                                |
| WWto2L2Nu_powheg                           | /WWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                  |
| WWto4Q_powheg                              | /WWto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                     |
| WWtoLnu2Q_powheg                           | /WWtoLnu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                  |
| WZto2L2Q_powheg                            | /WZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                                                   |

Continued on next page

Table 1: NanoAOD 2025 datasets used in the analysis (continued)

| Sample                              | NanoAOD dataset                                                                                                                                                   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WZto3LNu_powheg                     | /WZto3LNu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                  |
| WZtoLNu2Q_powheg                    | /WZtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                 |
| ZZto2L2Nu_powheg                    | /ZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                 |
| ZZto2L2Q_powheg                     | /ZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                  |
| ZZto2Nu2Q_powheg                    | /ZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                 |
| ZZto4L_powheg                       | /ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                    |
| WWW_4F                              | /WWW-4F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                  |
| WWZ_4F                              | /WWZ-4F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                  |
| WZZ                                 | /WZZ-5F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                  |
| ZZZ                                 | /ZZZ-5F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                                                  |
| ZHto2Tau_UncorrelatedDecay_Filtered | /ZH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v6/NANOADSIM |
| Zto2Nu_HT_100to200                  | /Zto2Nu-4Jets_Bin-HT-100to200_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                         |
| Zto2Nu_HT_200to400                  | /Zto2Nu-4Jets_Bin-HT-200to400_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                         |
| Zto2Nu_HT_400to800                  | /Zto2Nu-4Jets_Bin-HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                         |
| Zto2Nu_HT_800to1500                 | /Zto2Nu-4Jets_Bin-HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                        |
| Zto2Nu_HT_1500to2500                | /Zto2Nu-4Jets_Bin-HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                       |
| Zto2Nu_HT_2500toInf                 | /Zto2Nu-4Jets_Bin-HT-2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                             |
| Zto2Q_HT_100to400                   | /Zto2Q-4Jets_Bin-HT-100to400_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                          |
| Zto2Q_HT_400to800                   | /Zto2Q-4Jets_Bin-HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                          |
| Zto2Q_HT_800to1500                  | /Zto2Q-4Jets_Bin-HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                         |
| Zto2Q_HT_1500to2500                 | /Zto2Q-4Jets_Bin-HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM                        |
| Zto2Q_HT_2500toInf                  | /Zto2Q-4Jets_Bin-HT-2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM                              |